



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 3 - Day 2 Report

Internship Program — Batch 2026

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1. M-Tech Production & Marketing Internship Program 2026

Daily Internship Report

Date: 13 July 2026

2. 1. Introduction

This report summarizes the activities completed during the **M-Tech Production & Marketing AI/ML Internship Program** on **13 July 2026**. During today's session, I successfully studied and completed **Experiment 10: Computer Vision – Convolutional Neural Networks (CNNs) & OpenCV** from the **Artificial Intelligence & Machine Learning Lab Manual**. The experiment introduced the fundamentals of **Computer Vision** and explained how **Convolutional Neural Networks (CNNs)** and **OpenCV** are used for image processing, feature extraction, and image recognition tasks. Through this experiment, I gained practical knowledge of how computers analyze and interpret visual information to perform intelligent decision-making.

3. 2. Objectives

The primary objectives of today's learning session were:

- To understand the fundamental concepts of **Computer Vision**.
 - To study the architecture and working principles of **Convolutional Neural Networks (CNNs)**.
 - To explore image processing techniques using the **OpenCV** library.
 - To understand how convolution and pooling layers extract meaningful image features.
 - To learn how deep learning models classify and recognize images.
 - To explore real-world applications of Computer Vision in Artificial Intelligence.
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4. 3. Task Description

Today's task involved studying **Experiment 10** from the AI/ML Lab Manual, which focused on **Computer Vision** and the implementation of **Convolutional Neural Networks (CNNs)** for image classification.

The experiment introduced **OpenCV**, a widely used open-source computer vision library, and demonstrated how it can be used to load, display, resize, convert, and manipulate digital images. I also studied how **convolution layers**, **pooling layers**, and **fully connected layers** work together in CNNs to automatically extract visual features and classify images with high accuracy.

Additionally, I explored how Computer Vision technologies are applied in real-world Artificial Intelligence systems for image recognition, object detection, and visual data analysis.

5. 4. Work Done

During today's internship session, I successfully completed **Experiment 10** from the AI/ML Lab Manual.

I began by studying the fundamentals of **Computer Vision** and understanding how computers process and interpret visual information using Artificial Intelligence and Deep Learning techniques. I learned that Computer Vision enables machines to recognize objects, identify patterns, and make intelligent decisions based on images and videos.

Next, I explored the architecture of **Convolutional Neural Networks (CNNs)**. I studied the functions of **convolution layers**, **pooling layers**, **fully connected layers**, and the **output layer**, and learned how these components work together to extract meaningful visual features from images. I understood that convolution layers detect important patterns such as edges, textures, and shapes, while pooling layers reduce image dimensions without losing essential information, making the model more efficient.

I also gained practical knowledge of the **OpenCV** library and learned how it performs essential image processing operations such as reading images, resizing images, converting images to grayscale, and detecting important image features. These operations demonstrated how images can be prepared and enhanced before being used for Machine Learning and Deep Learning tasks.

Finally, I explored several real-world applications of Computer Vision, including **facial recognition**, **medical image analysis**, **autonomous vehicles**, **object detection**, **industrial automation**, and **security surveillance**. These practical examples highlighted the growing importance of Computer Vision technologies across multiple industries.

6. 5. Challenges Faced

The concepts presented in **Experiment 10** were engaging and practical. However, understanding how **convolution** and **pooling operations** mathematically extract image features required additional study. Learning how different CNN layers work together to identify increasingly complex visual patterns also required careful attention.

In addition, becoming familiar with the various functions available in the **OpenCV** library required practical experimentation. By reviewing examples, practicing image processing operations, and studying the complete CNN architecture, I developed a much stronger understanding of these concepts.

7. 6. Key Learnings

During today's learning session, I achieved the following outcomes:

- Developed a strong understanding of the fundamentals of **Computer Vision**.
- Learned the architecture and working principles of **Convolutional Neural Networks (CNNs)**.
- Understood the roles of **convolution**, **pooling**, **fully connected**, and **output** layers.
- Gained practical experience with basic image processing operations using the **OpenCV** library.
- Learned how CNNs automatically extract visual features for image recognition and classification.
- Understood the importance of image preprocessing in Deep Learning applications.
- Explored real-world applications of Computer Vision in **healthcare**, **security**, **robotics**, **autonomous vehicles**, **industrial automation**, and **Artificial Intelligence**.

8. 7. Conclusion

The successful completion of **Experiment 10** provided valuable theoretical knowledge and practical experience in **Computer Vision** and **Convolutional Neural Networks (CNNs)**. I developed a comprehensive understanding of how CNNs process images, extract meaningful features, and perform image classification with high accuracy. I also gained hands-on knowledge of using the **OpenCV** library for image preprocessing and manipulation.

The knowledge and practical skills acquired during today's session have significantly strengthened my understanding of image-based Artificial Intelligence systems and provided a solid foundation for future studies in advanced Computer Vision, Deep Learning, and AI applications. This learning will contribute greatly to my continued professional development throughout the internship program.

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10. Intern Information

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